

BEE 4750 Homework 1: Introduction to Using Julia

Name:

ID:

Due Date

Thursday, 9/11/25, 9:00pm

Overview

Instructions

- Problem 1 consist of a series of code snippets for you to interpret and debug. You will be asked to identify relevant error(s) and fix the code.
- Problem 2 asks you to write code to implement several simple functions using more general programming and some which are unique to Julia (which might mean that you need to look at the documentation or use Julia's help function, which is accessed with `?`, e.g. `?sum` for help with the `sum()` function) or to analyze Julia syntax.
- Problem 3 asks you to convert a verbal description of a wastewater treatment system into a Julia function, and then to use that function to explore the impact of different wastewater allocation strategies.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
Activating project at `c:\Users\holly\4750 HW\bee4750-hw1-ella-sandwich`
```

Standard Julia practice is to load all needed packages at the top of a file. If you need to load any additional packages in any assignments beyond those which are loaded by default, feel free to add a `using` statement, though [you may need to install the package](#).

```
using Random
using Plots
using GraphRecipes
using LaTeXStrings
using Distributions
```

```
# this sets a random seed, which ensures reproducibility of random number generation. You sh
Random.seed!(1)
```

```
TaskLocalRNG()
```

Problems (Total: 30 Points)

Problem 1 (12 points)

The following subproblems all involve code snippets that require debugging. You are encouraged to use online resources (*e.g.* Julia documentation and forums, Stack Overflow, Reddit, etc) to help find diagnose error messages and find solutions, but make sure that you clearly document which resources you used and how you used them.

Problem 1.1

You've been tasked with writing code to identify the minimum value in an array. You cannot use a predefined function. Your colleague suggested the function below, but it does not return the minimum value.

The initial minimum value at 0 was the issue, seeing as if you have all values in your array above 0, none will take the new value of the min. You have to start it as high as you can, which is why i chose infinity, and bring it down from there.

```
function minimum(array)
    # initialize the minimum value counter
    min_value = Inf64
    # update minimum values
    for i in 1:length(array)
```

```

        if array[i] < min_value
            min_value = array[i]
        end
    end
    # return found minimum
    return min_value
end

array_values = [89, 90, 95, 100, 100, 78, 99, 98, 100, 95]
@show minimum(array_values);

```

```
minimum(array_values) = 78
```

What is the logical flaw in the code? Fix it and show that your fixed code solves the problem.

Problem 1.2

Your team is trying to compute the average grade for your class, but the following code produces an `UndefVarError`.

The error means that the variable “average_grade” is undefined. The function was not called, so average_grade was not defined. After calling the function and setting it equal to “Average”, it takes the correct value.

```

# enter student grade vector
grades = [89, 90, 95, 100, 100, 78, 99, 98, 100, 95]
# compute class average

using Statistics
function class_average(student_grades)
    average_grade = mean(student_grades)
    return average_grade
end

Average = class_average(grades)

```

```
94.4
```

What does this `UndefVarError` mean? Why does this code produce one? Fix the error and solve the problem.

Problem 1.3

Your team wants to know the expected payout of an old Italian dice game called *passadieci* (which was analyzed by Galileo as one of the first examples of a rigorous study of probability). The goal of *passadieci* is to get at least an 11 from rolling three fair, six-sided dice. Your strategy is to compute the average wins from 1,000 trials, but the code you’ve written below produces a `MethodError`.

`MethodError` means that the wrong arguments were given to a function, built-in or created by you. I redid some of the function organization so it made more sense to me, and discovered that the “outcomes” array was incorrectly made. Changing it to read “`zeros(n_trials)`” creates an array of 0s that can be changed depending on win/loss.

```
# function to simulate a passadieci roll: 3 6-sided dice
function passadieci()
    # this rand() call samples 3 values from the vector [1, 6]
    roll = rand(1:6, 3)
    return roll
end
# set number of trials and initialize outcome vector
n_trials = 1_000
outcomes = zeros(n_trials)
# simulate number of passadieci rolls and count wins
for i = 1:n_trials
    sum_roll = sum(passadieci())
    if sum_roll >= 11
        outcomes[i] = 1
    #else
    #    outcomes[i] = 0
    end
end
win_prob = sum(outcomes) / n_trials # compute average number of wins
@show win_prob;
```

```
win_prob = 0.492
```

What does this `MethodError` mean? Why does this code produce one? Fix the error and solve the problem.

Problem 1.4

You're interested in writing some code to remove the mean of a vector from all of its components. You've written the following code and tried to test it on a random vector, but your code returns a `MethodError`.

This current code is attempting to subtract one value from a vector, so the method does not make sense and it doesn't match up. You have to instead go through the entire vector and individually subtract from each index. I programmed my function to do just that, and the error message went away. When testing on different vectors, there is sometimes an error regarding repeating decimals (3.33333...), but after rounding the mean the message disappeared and the code worked perfectly.

```
# function to remove mean from a vector
function remove_mean(vect)
    # function to compute the mean
    function compute_mean(vect)
        element_sum = 0 # initialize sum
        # compute mean and return
        for v in vect
            element_sum += v
        end
        return element_sum / length(vect)
    end

    m = (compute_mean(vect)) # compute mean
    # return demeaned vector
    for i in 1:length(vect)
        vect[i] = vect[i] - m
    end
    return vect
end

random_vect = rand(1_000)
@show remove_mean(random_vect)
```

```
remove_mean(random_vect) = [0.37317987893195814, -0.06289375848122813, -0.16925379463472623,
```

```
1000-element Vector{Float64}:
 0.37317987893195814
-0.06289375848122813
-0.16925379463472623
 0.3167102214503681
-0.009825368803801915
```

```

-0.25376600997075616
-0.2201098293177176
-0.39012691419794787
-0.05558350407567336
-0.2733029743084483

-0.44678376394334096
 0.22371617726689952
-0.13397347221412714
-0.34052540344364945
-0.4205215650310934
-0.1382104975688474
-0.09287919547835377
-0.4848112427361262
-0.4384237932117363

```

What does this `MethodError` mean? Why does this code produce one? Fix the error and solve the problem.

Problem 2 (18 points)

Cheap Plastic Products, Inc. is operating a plant that produces $100\text{m}^3/\text{day}$ of wastewater that is discharged into Pristine Brook. The wastewater contains $1\text{kg}/\text{m}^3$ of YUK, a toxic substance. The US Environmental Protection Agency has imposed an effluent standard on the plant prohibiting discharge of more than $20\text{kg}/\text{day}$ of YUK into Pristine Brook.

Cheap Plastic Products has analyzed two methods for reducing its discharges of YUK. Method 1 is land disposal, which costs $X_1^2/20$ dollars per day, where X_1 is the amount of wastewater disposed of on the land (m^3/day). With this method, 20% of the YUK applied to the land will eventually drain into the stream (*i.e.*, 80% of the YUK is removed by the soil).

Method 2 is a chemical treatment procedure which costs $\$1.50$ per m^3 of wastewater treated. The chemical treatment has an efficiency of $e = 1 - 0.005X_2$, where X_2 is the quantity of wastewater (m^3/day) treated. For example, if $X_2 = 50\text{m}^3/\text{day}$, then $e = 1 - 0.005(50) = 0.75$, so that 75% of the YUK is removed.

Cheap Plastic Products is wondering how to allocate their wastewater between these three disposal and treatment methods (land disposal, chemical treatment, and direct disposal) to meet the effluent standard while keeping costs manageable.

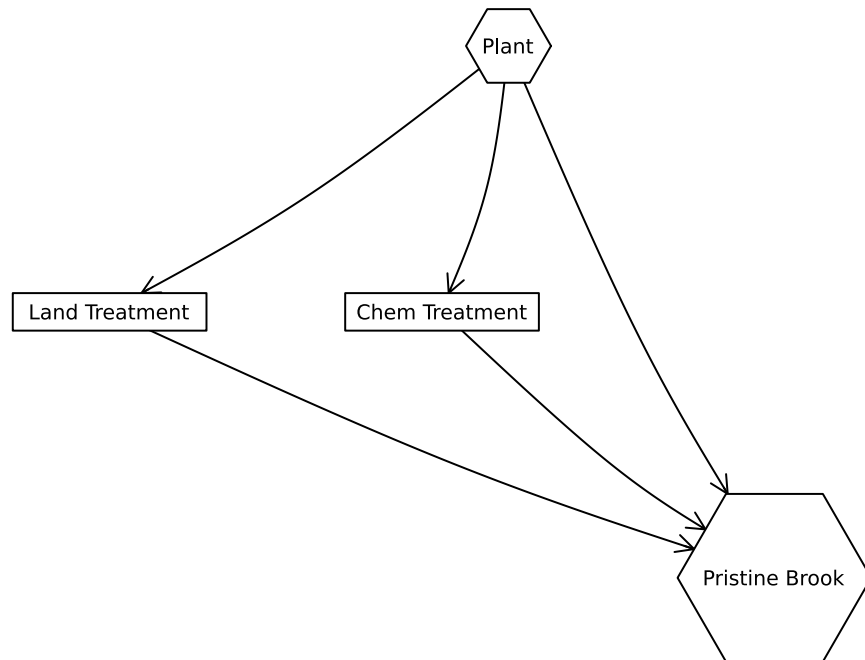
The flow of wastewater through this treatment system is shown in Figure 1. Modify the edge labels (by editing the `edge_labels` dictionary in the code producing Figure 1) to show how the wastewater allocations result in the final YUK discharge into Pristine Brook. For the

`edge_label` dictionary, the tuple (i, j) corresponds to the arrow going from node i to node j . The syntax for any entry is $(i, j) \Rightarrow \text{"label text"}$, and the label text can include mathematical notation if the string is prefaced with an L, as in L"x₁" will produce x_1 .

```
A = [0 1 1 1;
      0 0 0 1;
      0 0 0 1;
      0 0 0 0]

names = ["Plant", "Land Treatment", "Chem Treatment", "Pristine Brook"]
# modify this dictionary to add labels
edge_labels = Dict((1, 2) => "", (1, 3) => "", (1, 4) => "", (2, 4) => "", (3, 4) => "")
shapes=[:hexagon, :rect, :rect, :hexagon]
xpos = [0, -1.5, -0.25, 1]
ypos = [1, 0, 0, -1]

p = graphplot(A, names=names, edglabel=edge_labels, markersize=0.15, markershapes=shapes, ma
display(p)
```



There will be 3 equations coupled depending on X1, X2, and X3 (the discharge reaching each

treatment), where resulting deposits in the brook are Y_1 , Y_2 , and Y_3 , summing together to result in the total deposit which must not exceed 20 kg/day 1: $Y_1 = X_1 * 0.2$; 2: $Y_2 = X_2 - X_2 * (1 - 0.005X_2)$; 3: $Y_3 = X_3 \text{ 1kg/m}^3$; Assumptions: ($20 \geq Y_1 + Y_2 + Y_3$), and ($X_1 + X_2 + X_3 = 100$)

As for cost: 1: $C_1 = X_1^2/20$; 2: $C_2 = X_2 * 1.5$; 3: $C_3 = \text{free}$; Total cost = $C_1 + C_2 + C_3$;

Problem 2.1

Formulate a mathematical model for the treatment cost and the amount of YUK that will be discharged into Pristine Brook based on the wastewater allocations. This is best done with some equations and supporting text explaining the derivation. Make sure you include, as additional equations in the model, any needed constraints on relevant values. You can find some basics on writing mathematical equations using the LaTeX typesetting syntax [here](#), and a cheatsheet with LaTeX commands can be found on the course website's [Resources page](#).

Problem 2.2

Implement your systems model as a Julia function which computes the resulting YUK discharge and cost for a particular treatment plan. You can return multiple values from a function with a [tuple](#), as in:

I created my function to take ratios for different treatment plans and convert those into actual outputs, including resulting deposited amount and resulting cost. As long as the ratios add up to less than 1.1 (not 1 due to rounding issues from certain numbers), the code will produce the resulting cost and deposited amount.

```
function yuk_discharge(x1, x2, x3)
    if x1 + x2 + x3 > 1.1
        print("Incorrect Ratios")
    else
        # amount that reaches treatment facility based on given ratio
        X1 = x1 * 100
        X2 = x2 * 100
        X3 = x3 * 100
        # deposited amount in pristine brook
        Y1 = X1 * 0.2
        Y2 = X2 - (X2 * (1 - (0.005 * X2)))
        Y3 = X3 * 1
        # total deposited amount
        Y = Y1 + Y2 + Y3
        # cost
        C1 = (X1^2)/20
```



```

        C2 = X2 * 1.5
        C3 = 0
        # total cost
        C = C1 + C2 + C3
    return (Y, C)
end
end

```

yuk_discharge (generic function with 1 method)

```

# Function Test
Y, C = yuk_discharge(0.8, 0.2, 0)

```

(18.0, 350.0)

```

function multiple_return_values(x, y)
    return (x+y, x*y)
end

```

```

a, b = multiple_return_values(2, 5)
@show a;
@show b;

```

```

a = 7
b = 10

```

To evaluate the function over vectors of inputs, you can *broadcast* the function by adding a decimal `.` before the function arguments and accessing the resulting values by writing a *comprehension* to loop over the individual outputs in the vector:

```

x = [1, 2, 3, 4, 5]
y = [6, 7, 8, 9, 10]

output = multiple_return_values.(x, y)
a = [out[1] for out in output]
b = [out[2] for out in output]
@show a;
@show b;

```

```
a = [7, 9, 11, 13, 15]
b = [6, 14, 24, 36, 50]
```

2.3: In order to evaluate 1000 combinations, I created a matrix with 1000 columns and 3 rows, and normalized them so the columns add up to 1. I then converted this matrix into 3 row-vectors, and evaluated all 1000 at the same time using the broadcast feature explained above. There are a few combinations that are in compliance with the 20kg/day limit, but these are quite expensive. The cheapest one I see is around \$275. As expected, the cheaper treatments result in more YUK discharge, and even the most expensive treatments may not satisfy the requirements. In order to find a better solution, I would like to have a budget, and maybe the importance of lowering the discharge numbers past the legal value. The ranking of importance between cost and discharge amount would further hone in the combinations we should use. For example, if a lower YUK discharge is valued and cost is not a constraint, the combination with the lowest YUK discharge would win, no matter the cost.

2.4: In order to minimize cost, we should go fully with the cheapest option if YUK discharge is not considered. In this case, direct disposal is free and therefore cheaper than any other treatment option. This will result in a cost of \$0, but a YUK discharge of $100\text{m}^3/\text{day} * 1\text{kg}/\text{m}^3 = 100\text{ kg}/\text{day}$ of YUK. As for minimizing YUK, we should only use the two options that actually remove the chemical (option 1 and option 2). Using the 2 equations we have for YUK disposal into the brook and the constraints $Y1+Y2<20$ and $X1+X2=100$, I combined the 2 equations to set an equation in terms in $X2$ equal to $Y1+Y2$, then finding the minimum of that equation, I found that $X2=20$ and therefore $X1=80$. This results in a deposited amount of 18 kg/day, which is consistent with the graph above (nothing above is less than 18). This solution will cost \$350. These two points follow the pattern shown by the previous scatterplot, with the minimum YUK (18kg) being fairly close to other points of other combinations. However, there are other points on the plot that fall below the red dotted line that are cheaper. I would most likely go with these if I were on a budget, seeing as the deposited amount is still close to the minimum value of 18kg. As for the minimum cost, it is a little more of an outlier, but still follows the pattern. I would certainly not go with this one, seeing as it serves no environmental benefit.

```
# 2.3
# random matrix generator
M = rand(3, 1000)
# make columns add up to 1
column_sums = sum(M, dims=1)
norm_M = M ./ column_sums
# create 3 vectors, which are the rows in matrix M
A = norm_M[1,:]
B = norm_M[2,:]
C = norm_M[3,:]
# Find output and Cost for each combo (x1000)
```

```

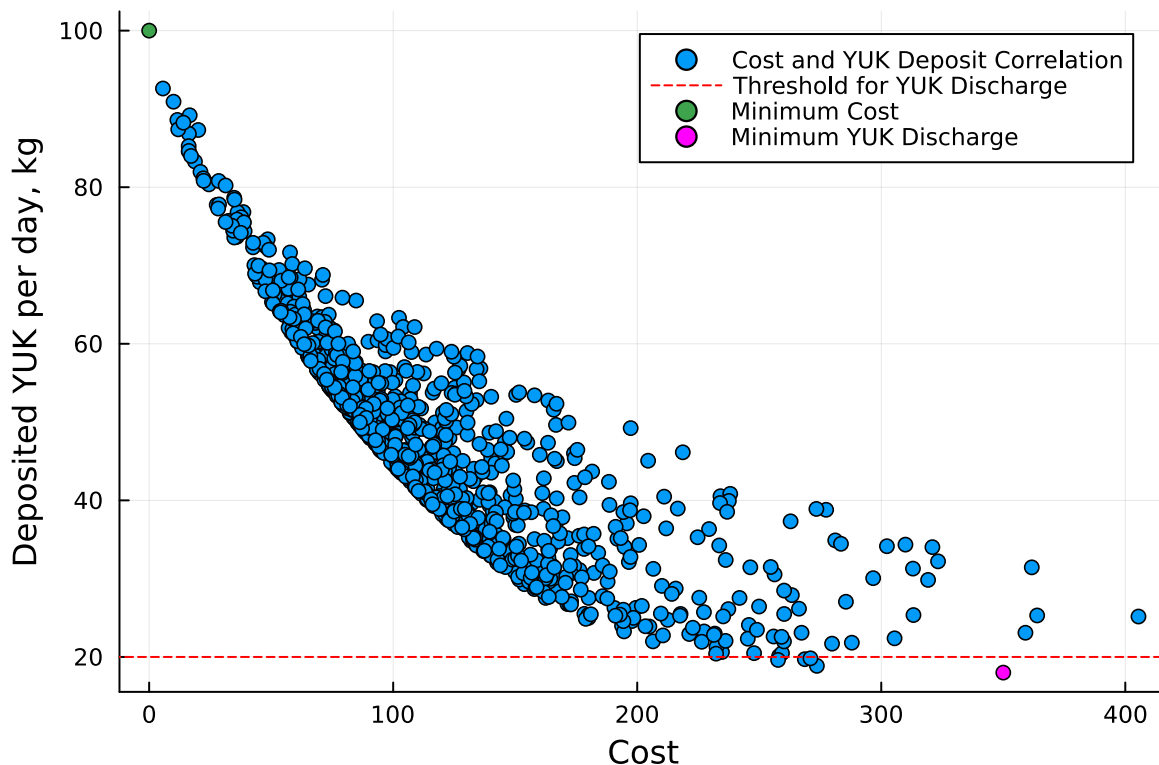
output = yuk_discharge.(A, B, C)
Y = [out[1] for out in output];
C = [out[2] for out in output];
@show Y
@show C

# Plot the results, with a line at y=20

import Pkg;
Pkg.add("Plots")
plot(C, Y, seriestype=:scatter, xlabel = "Cost", ylabel = "Deposited YUK per day, kg", label=:none)
hline!(C, [20], linecolor=:red, linestyle=:dash, label="Threshold for YUK Discharge")

# Plot 2.4 values
min_cost_point = (0, 100)
min_yuk_point = (350, 18)
scatter!(min_cost_point, label="Minimum Cost")
scatter!(min_yuk_point, color=:magenta, label="Minimum YUK Discharge")

```



Make sure you comment your code appropriately to make it clear what is going on and why.

Problem 2.3

Use your function to experiment with 1,000 different combinations of wastewater discharge and treatment. You can do this with either a grid search or by sampling from a [Dirichlet distribution](#) (a $\text{Dirichlet}(1, n)$ distribution will generate uniformly-weighted n -dimensional vectors whose components add up to 1; see the [Distributions.jl documentation](#) for how to sample from probability distributions in Julia). Plot the results of these experiments. Do any satisfy the YUK effluent standard (plot this as well as a dashed red line). What was the cost of solutions satisfying the standard? What can you say about the tradeoff between treatment cost and YUK discharge? You don't have to find an "optimal" solution to this problem, but what do you think would be needed to find a better solution?

Problem 2.4

Find the strategies which minimize cost and YUK discharge (these will be different strategies) analytically and find the values of the objective metrics. Plot these values in the plot that you created for Problem 2.3. How do their values compare to the spread of values that you found in that problem? Would you select either of them (explain why or why not)?

References

List any external references consulted, including classmates.